INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

SIMULATOR, PROJECTILE, GROUND BURST – M115A2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the parts, assembly and verification methods for the Simulator, Projectile, Ground Burst: M115A2.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: RDAR-EIQ-SA, Picatinny Arsenal, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online at https://assist.dla.mil .

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-12597	-	Igniter, Blasting fuse, friction type, M3A1 Loading, Assembly and Packing
MIL-P-48239	-	Powder, Flash

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-331	-	Fuse and fuse Components, Environmental and Performance Tests For
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available on line at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC)

PRODUCT DRAWINGS

7549218	-	Sleeve
7549227	-	Whistle Loading Assembly
7549229	-	Outer Tube
7549232	-	Spacer Tube
7549233	-	Cover Assembly
7549234	-	Inner Tube

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7549235	-	Whistle and Cover Assembly
7549246	-	Simulator, Projectile, Ground Burst, M115A2, Assembly
8833721	-	Igniter, Blasting Fuse, Friction Type, M3A1 Assembly
8865588	-	Charge Loading Assembly
9345134	-	Safety Fuse Cord
9345136	-	Charge Loading Assemble, Alternate
9345140	-	Whistle and Cover Assembly, Alternate
12972326	-	Flash Composition
13019989	-	Flash Composition
13020526	-	Charge Loading Assembly

INSPECTION DRAWINGS

9200454	-	Launcher

(Copies of these drawings maybe requested online at usarmy.pica.rdecom-ardec.list.drawing-request-help-desk@mail.mil or from US Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NON-GOVERNMENT STANDARDS

ISO 10070	-	Determination of envelope-specific surface area From measurements of the permeability to air of a Powder bed under steady-state flow conditions

(Copies of this document are available from www.iso.org or International Organization for Standardization Case Postale 56, CH-1211 Geneve 20, Switzerland)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the even of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirement inspections.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection in accordance with 4.2.

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3.1.2 Conformance. All materials, components and assemblies shall be in accordance with the requirements in the drawing listed in paragraphs 4.3.2.1 – 4.3.2.13.

3.2 Components and assemblies. All materials, components and assemblies shall be in accordance with the requirements on the drawing listed in paragraphs 4.3.2.1 – 4.3.2.13.

3.2.1 Inner tube. The inner tube material and dimensions shall be in accordance with 7549234 .

3.2.2 Outer tube. The outer tube material and dimensions shall be in accordance with 7549229.

3.2.3 Sleeve. The sleeve material and dimensions shall be in accordance with 7549218.

3.2.4 Spacer tube. The spacer tube material and dimensions shall be in accordance with 7549232.

3.2.5 Whistle loading assembly. The whistle loading assemble shall be in accordance with 7549227.

3.2.6 Cover assembly. The cover assembly shall be in accordance with 7549233.

3.2.7 Whistle and cover assembly. The whistle and cover assembly shall be in accordance with 7549235.

3.2.8 Charge loading assembly, prior to sealing cup assembly. The charge loading assembly, prior to sealing cup assembly shall be assembled in accordance with 12972326 or 13019989.

3.2.9 Charge loading assembly (BP-A1). The cover and body loading assembly for part number 7549246-1, NSN 1370-01-557-8527 shall be in accordance with 8865588.

3.2.10 Charge loading assembly (A1-KP). The cover and body loading assembly for part number 7549246-2, NSN 1370-00-752-8126 shall be in accordance with 13020526.

3.2.11 Assembly, prior to assembly of disc and washer assembly. The assembly, prior to assembly of disc and washer assembly shall be assembled in accordance with 7549246.

3.2.12 Assembly. The assembly shall be in accordance with 7549246.

3.2.12.1 Outer tube wrinkles. The outer tube shall be free of cuts, holes, tears or wrinkles.

3.2.13 M3A1 igniter assembly – The M3A1 igniter assembly shall be in accordance with 8833721.

3.2.14 Safety fuse cord. The fuse assembly material and dimensions shall be in accordance with 9345134.

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3.2.14.1 Burning time of safety fuse. Each safety fuse, prepared for assembly to the simulator, shall be cut perpendicular to its axis without fraying or damaging the safety fuse. It shall burn within the time limits specified on the applicable drawing following bending to simulate the bend from the lasting fuse igniter to the whistle loading assembly of the simulator. The length of the fuse cord shall be cut to determine a burn time of the safety fuse to be greater than or equal to six (6) and less than or equal to ten (10) seconds. A burn time less than five (5) is considered a critical defect. A burn time equal time greater than or equal to five (5) seconds and less than six (6) seconds is considered a major defect.

3.3 Moisture content

3.3.1 Flash composition moisture content. The moisture content for the flash composition for part number 7549246-1, (NSN 1370-01-557-8527) shall be in accordance with 8865588 and shall be in accordance with 13020526 for part number 7549246-2 (NSN 1370-00-752-8126).

3.3.2 Whistle composition moisture content. Whistle composition shall be in accordance with 7549227 and the moisture content of the whistle composition at the loading station at the time of loading shall not exceed 0.10 percent.

3.3.3 Flash composition formulation. The formulation for the flash composition for part number 7549246-1, (NSN-1370-01-557-8527) shall be in accordance with 12972326 and shall be in accordance with 13019989 for part number 7549246-2 (NSN 1370-00-752-8126).

3.3.4 Flash composition granulation. The granulation for the flash composition for part number 7549246-1, (NSN 1370-01-557-8527) shall be in accordance with 12972326 and shall be in accordance with 13019989 for part number 7549246-2 (NSN 1370-00-752-8126).

3.4 Jumble. The simulator shall comply with the following requirements when subjected to a jumble environment:

a. The simulator shall not function during test.
b. There shall be no loose pyrotechnic powder on the simulator or in the test box after jumbling.
c. The simulator components shall not separate during the test.

3.5 Functioning requirements. The simulators shall function at ambient (70° ± 5° F) and cold temperatures (minus 65 ±5° F) and shall comply with the following requirements:

3.5.1 Premature delay. The simulator shall not function prematurely (less than five (5) seconds delay) and the delay time shall be greater or equal to six (6) seconds and less than or equal to ten (10) seconds. After safety fuse burns for six (6) to ten (10) seconds, the simulator shall whistle for three (3) to six (6) seconds before the simulator explodes.

3.5.2 Sound level intensity. The sound level intensity shall be not less than 138 decibels at a distance of 75 ± 0.5 feet from the simulator.

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3.5.3 Pull cord. The pull cord shall not break or otherwise separate from the igniter wire when used to function the simulator.

3.5.4 Audible whistle. The assembly shall produce an audible whistle for the required time as specified on drawing 7549227.

3.6 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.7 Workmanship. All parts and assemblies shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpage, burn marks, checks, chipped edges, blisters, excess flash, dirt, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part nor shall the cleaning agent contaminate parts.

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4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Method of Verification	Classes of Verification
1- ANALYSIS 2- DEMONSTRATION (end item test) 3- EXAMINATION 4- TEST	A – FIRST ARTICLE B – CONFORMANCE INSPECTION

Section 3 Requirements	Description	Verification Method				Verification Class		Section 4
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2 and TABLE II
3.1.2	Conformance	X		X	X		X	4.3 and 4.3.2
3.2.1	Inner tube			X	X	X	X	4.3.2.1
3.2.2	Outer tube			X	X	X	X	4.3.2.2
3.2.3	Sleeve			X	X	X	X	4.3.2.3
3.2.4	Spacer tube			X	X	X	X	4.3.2.4
3.2.5	Whistle loading assembly			X	X	X	X	4.3.2.5
3.2.6	Cover assembly			X	X	X	X	4.3.2.7
3.2.7	Whistle and cover assembly			X	X	X	X	4.3.2.8
3.2.8	Charge loading assembly, prior to sealing cup assembly			X	X	X	X	4.3.2.9
3.2.9	Charge loading assembly (BP-A1)			X	X	X	X	4.3.2.10
3.2.10	Charge loading assembly (A1-KP)			X	X	X	X	4.3.2.11
3.2.11	Assembly, prior to assembly of disc and washer assembly			X	X	X	X	4.3.2.12
3.2.12	Assembly			X		X	X	4.3.2.13
3.2.13	M3A1 igniter assembly			X	X	X	X	
3.2.14	Safety fuse cord			X	X	X	X	4.3.2.6
3.2.14.1	Burning time of safety fuse			X	X	X	X	4.3.2.6
3.3	Moisture content	X		X	X	X	X	4.7.1
3.3.1	Flash composition moisture content	X		X	X	X	X	4.7.1
3.3.2	Whistle composition moisture content	X		X	X	X	X	4.7.1
3.3.3	Flash composition formulation	X		X	X	X	X	4.7.2
3.3.4	Flash composition granulation	X		X	X	X	X	4.7.2
3.4	Jumble			X	X	X		4.7.3
3.5	Functioning requirements			X	X	X	X	4.5.3
3.5.1	Premature delay			X	X	X	X	4.5.3
3.5.2	Sound level intensity			X	X	X	X	4.5.3
3.5.3	Pull cord			X	X	X	X	4.5.3

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TABLE I. Requirement/verification cross reference matrix-Continued

3.5.4	Audible whistle			X	X	X	X	4.5.3
3.6	Lot numbering			X		X	X	4.8
3.7	Workmanship			X		X	X	4.3.2

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows.

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article. When specified, a sample shall be subjected to first article inspection in accordance with Table II.

4.2.1 First article rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, first article requirements, the first article sample shall be rejected.

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TABLE II. First article inspection.

Examination or Test	Drawing number	No. of samples	Requirement Paragraph	Inspection Method/ Reference
Inner tube Examination for defects	7549234	25	3.2.1	4.3.2.1
Outer tube Examination for defects	7549229	25	3.2.2	4.3.2.2
Sleeve Examination for defects	7549218	25	3.2.3	4.3.2.3
Spacer tube Examination for defects	7549232	25	3.2.4	4.3.2.4
Whistle loading assembly Examination for defects	7549227	25	3.2.5	4.3.2.5
Cover assembly Examination for defects	7549233	25	3.2.6	4.3.2.7
Whistle and cover assembly Examination for defects	7549235	25	3.2.7	4.3.2.8
Charge loading assembly, prior to sealing cup assembly Examination for defects	12972326 or 13019989	25	3.2.8	4.3.2.9
Charge loading assembly (BP-Al) Examination for defects	8865588	25	3.2.9	4.3.2.10
Charge loading assembly (Al-KP) Examination for defects	13020526	25	3.2.10	4.3.2.10
Assembly, prior to assembly of disc and washer assembly Examination for defects	7549246	25	3.2.11	4.3.2.11
Assembly Examination for defects	7549246	25	3.2.12	4.3.2.12
M3A1 igniter assembly Examination for defects	8833721	Note a.	3.2.13	Note a.
Safety fuse cord Examination for defects	9345134	25	3.2.14	4.3.2.6

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TABLE II. First article inspection - Continued

Moisture content (7549246-1)				
Flash composition	8865588	Note d.	3.3.1	4.5.1 / 4.7.1
Whistle composition	7549235	Note d.	3.3.2	4.5.1 / 4.7.1
Flash composition formulation	12972326	Note e.	3.3.3	4.5.2 / 4.7.2
Flash composition granulation	12972326	Note e.	3.3.4	4.5.2 / 4.7.2
Moisture content (7549246-2)				
Flash composition	13020526	Note d.	3.3.1	4.5.1 / 4.7.1
Whistle composition	7549235	Note d.	3.3.2	4.5.1 / 4.7.1
Flash composition formulation	13019989	Note e.	3.3.3	4.5.2 / 4.7.2
Flash composition granulation	13019989	Note e.	3.3.4	4.5.2 / 4.7.2
Simulator (complete assembly)				
Examination for defects	7549246	50	3.2.11	4.3.2.12
Jumble		10	3.4	4.7.3
Functioning – ambient @ 65°F to 75°F		20 (Note b)	3.5	4.5.3
Functioning – cold @ -70°F to -60°F		10 (Note c)	3.5	4.5.3

Notes:
a. Test and acceptance in accordance with first article requirements of MIL-I-12597.
b. Ten assemblies functioned in static (4.7.4.1.1) and non- static phase (4.7.4.1.2).
c. Ten assemblies functioned in the non- static phase (4.7.4.1.2).
d. Test and acceptance in accordance with first article requirements from paragraphs 4.5.1 and 4.7.1.
e. Test and acceptance in accordance with first article requirements from paragraphs 4.5.2 and 4.7.2.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lot is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance of applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revision, same specifications and same specification revisions and complies with provisions for submission of product as a specified in MIL-STD-1916, paragraph "Acceptance by tables."

4.3.1.1 Lot numbering. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. In addition, inspection lots of loaded simulators shall contain:

a. Parts from one lot interfix from one manufacturer.
b. Safety fuse from not more than one lot interfix number from one manufacturer.
c. Each ingredient of the flash and whistle composition of one lot interfix number from not more than one manufacturer.

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d. M3A1 Fuse igniters from not more than one lot interfix number from one manufacturer.

4.3.2 Classification of Characteristics. For the conformance inspection paragraphs of 4.3.2.1 to 4.3.2.13, the definitions of critical, major and minor defects are in accordance with MIL-STD-1916 (also see 6.5 for critical characteristic justification). Unless otherwise specified, sampling plans and acceptance criteria for critical, major and minor characteristics shall be in accordance with MIL-STD-1916.

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4.3.2.1	Inner tube.			DRAWING NUMBER 7549234
				NEXT HIGHER ASSEMBLY 7549233
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major	None defined			
Minor				
201	Length	VL-II	3.2.1	Gage
202	Inside diameter	VL-II	3.2.1	Gage
203	Evidence of poor workmanship	VL-II	3.7	Visual
NOTES:				

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4.3.2.2	Outer tube.			DRAWING NUMBER 7549229
				NEXT HIGHER ASSEMBLY 7549233
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Length	VL-IV	3.2.2	Gage
102	Wall thickness (Note a)	VL-IV	3.2.2	Gage
Minor				
201	Inside diameter	VL-II	3.2.2	Gage
202	Evidence of poor workmanship	VL-II	3.7	Visual

NOTES:
a. Wall thickness shall be gaged in non-overlap regions only.

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4.3.2.3	Sleeve.			DRAWING NUMBER 7549218 NEXT HIGHER ASSEMBLY 7549227
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Location of hole	VL-IV	3.2.3	Gage
102	Sleeve snug slide fit over whistle tube	VL-IV	3.2.3	Manual (note a)
Minor				
201	Evidence of poor workmanship	VL-II	3.7	Visual

NOTES:

a. The part shall slide snuggly fit over the whistle tube (dwg. 7549219) with applying slight but not excessive force to fit over the whistle tube. Part 9549218 shall randomly be paired with part 7549219 to verify that all parts are snug slide fit over the whistle tube.

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4.3.2.4	Spacer tube.			DRAWING NUMBER 7549232
				NEXT HIGHER ASSEMBLY 7549233
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major	None defined			
Minor 201	Evidence of poor workmanship	VL-II	3.7	Visual
NOTES:				

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4.3.2.5	Whistle loading assembly.			DRAWING NUMBER 7549227
				NEXT HIGHER ASSEMBLY 7549235
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major 101	Charge missing	VL-IV	3.2.5	Visual
Minor 201	Evidence of poor workmanship	VL-II	3.7	Visual
NOTES:				

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4.3.2.6	Fuse cord assembly.			DRAWING NUMBER 9345134
				NEXT HIGHER ASSEMBLY 7549246
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	Short fuse delay (less than five (5) seconds)	100%	3.2.14	4.6.1.1
Major				
101	Short length	VL-IV	3.2.14	4.6.1.1
102	Fuse delay greater than ten (10) seconds	VL-IV	3.2..14	4.6.1.1
103	Safety fuse frayed or damaged	VL-IV	3.2.14	4.6.1.1
Minor				
201	Evidence of poor workmanship	VL-II	3.7	Visual
NOTES:				

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4.3.2.7	Cover assembly.			DRAWING NUMBER 7549233
				NEXT HIGHER ASSEMBLY 7549235
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Any component missing	VL-IV	3.2.6	Visual
102	Contact surfaces and joints not sealed securely	VL-IV	3.2.6	Visual & Manual
Minor				
201	Total length	VL-II	3.2.6	Gage
202	Evidence of poor workmanship	VL-II	3.7	Visual
NOTES:				

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4.3.2.8	Whistle and cover assembly.			DRAWING NUMBER 7549235
				NEXT HIGHER ASSEMBLY 7549246
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Insufficient adhesive between tubes	VL-IV	3.2.7	Visual
102	Any component missing	VL-IV	3.27	Visual
103	Whistle assembly insecure	VL-IV	3.27	Manual
Minor				
201	Evidence of poor workmanship	VL-II	3.7	
NOTES:				

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4.3.2.9	Charge loading assembly, prior to sealing cup assembly.			DRAWING NUMBER 12972326 or 13019989
				NEXT HIGHER ASSEMBLY 7549246
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major 101	Weight of charge	Note a	3.2.8	Balance
Minor 201	Evidence of poor workmanship	VL-II	3.7	Visual

NOTES:

a. A sample of fifty (50) charge loading assemblies, prior to sealing the cup assembly shall be selected from each four (4) hours of production and subjected to a weight of charge verification as follows:

Using a tared whistle and cover assembly (dwg. 7549235), the weight difference due to the addition of the flash composition is determined. If the weight of the charge fails to meet the applicable requirements, those assemblies loaded since the last acceptable verification test shall be classed defective and removed from the lot. (non-destructive test).

Alternate Method: Empty the contents of the charge loading assembly (dwg. 8865588) into a tared pan and determine the weight of the flash composition.

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4.3.2.10	Charge loading assembly (BP-Al) NSN 1370-01-557-8527.			DRAWING NUMBER 8865588
				NEXT HIGHER ASSEMBLY 7549246-1
	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Assembly damaged to extent that powder may leak	VL-IV	3.2.9	Visual
102	Assembly loose at any joint or contact	VL-IV	3.2.9	Visual & Manual
Minor				
201	Evidence of poor workmanship	VL-II	3.7	Visual
NOTES:				

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4.3.2.11	Charge loading assembly (Al-KP) NSN 1370-00-752-8126.			DRAWING NUMBER 13020526
				NEXT HIGHER ASSEMBLY 7549246-2
	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Assembly damaged to extent that powder may leak	VL-IV	3.2.10	Visual
102	Assembly loose at any joint or contact	VL-IV	3.2.10	Visual & Manual
Minor				
201	Evidence of poor workmanship	VL-II	3.7	Visual
NOTES:				

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4.3.2.12	Assembly, prior to assembly of disc and washer assembly.			DRAWING NUMBER 7549246
				NEXT HIGHER ASSEMBLY 7549246
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major 101	Quick match insecure or incorrectly loaded	VL-IV	3.2.11	Visual & Manual
Minor 201	Evidence of poor workmanship	VL-II	3.7	Visual
NOTES:				

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4.3.2.13	Assembly.			DRAWING NUMBER 7549246 NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	Any opening in assembly through which charge may leak, or presence of charge on exterior of assembly	100%	3.2.12	Visual
2	Safety clip missing, insecurely engaged or incorrectly positioned	100%	3.2.12	Visual
3	Premature functioning (time from initiation to detonation less than 5 seconds)	Note a	3.5	4.5.3, 4.7.4
Major				
101	Fuse tape or sealing tape missing, peeling or incorrectly applied	VL-IV	3.2.12	Visual
102	Vent hole missing in fuse tape	VL-IV	3.2.12	Visual
103	Cement missing or inadequate at safety fuse or disc contacting surface (as applicable)	VL-IV	3.2.12	Visual
104	Paint on igniter blasting fuse will interfere with removal of cap and pull cord	VL-IV	3.2.12	Visual
105	Assembly cut, punctured, dented or deformed	VL-IV	3.2.12	Visual
106	Total length	VL-IV	3.2.12	Gage
107	Label missing or insecurely attached	VL-IV	3.2.12	Visual & Manual
108	Delay time (safety fuse delay between (5) to six (6) seconds – DWG 7549246	Note a	3.5	4.6.3

NOTES:
Note a. Sample size shall be selected in accordance with 4.5.3

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4.3.2.13	Assembly (continued).			DRAWING NUMBER 7549246 NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Major				
109	Delay time (safety fuse delay between five (5) to six (6) seconds – DWG 7549246)	Note a	3.5	4.6.3
110	Delay time (safety fuse delay over maximum – DWG 8549246)	Note a	3.5	4.6.3
111	Functioning (No audible whistle)	Note a	3.5	4.5.3
112	Functioning (sound level intensity less than minimum)	Note a	3.5	4.5.3
113	Functioning (assembly does not function – 'dud")	Note a	3.5	4.5.3
114	Functioning (Pull cord breaks or separates)	Note a	3.5	4.5.3
Minor				
201	Functioning (whistle duration incorrect)	Note a	3.5	4.7.1.1
202	Tube Wrinkled	VL-II	3.2.12.1	Visual
203	Poor workmanship	VL-II	3.7	Visual

NOTES:

a. Sample size shall be selected in accordance with 4.5.3

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4.5 Testing.

4.5.1 Moisture content

TABLE III. Major defects for moisture content.

Material	Drawing	Inspection Method Paragraph
Whistle	7549227	4.7.1.1
Flash composition or ingredients	8865588 or 13020526	4.7.1.2

4.5.1.1 Material samples moisture content. For verification, one sample of each material as representative of eight (8) hours of production of simulators shall be selected and subject to test. Composite samples shall not be used. If the moisture content of the sample exceeds the requirement, that quantity or sub-lot of material represented by the sample shall not be used in production. If the quantity of material or sub-lot with excessive moisture has been used in loading and packing, the remaining unloaded and unpacked material shall not be used in production, and the loaded and packed simulators shall be rejected.

4.5.1.2 Simulator ingredients moisture content. A sample of sufficient size (see paragraph 4.7.1) from each lot of each ingredient at the time of inserting the ingredient to the simulator for determination of moisture content shall be selected. Controls shall be provided so that the moisture content is maintained under constant temperature and humidity by a graphic recorder up to and at the time of sealing and blending the charge in the simulators. If the sample fails to meet the requirements, or if the simulators have been loaded with ingredients containing excessive moisture, the remainder of the ingredient lot shall not be used in further production, and the lot of loaded simulators shall be rejected.

4.5.2 Determination of granulation and composition of flash powder – Major defect.

TABLE IV. Major defects for granulation and composition of flash powder.

Material	Drawing	Inspection Method Paragraph
Granulation	12972326 or 13019989	4.7.2.1
Composition	12972326 or 13019989	4.7.2.2

4.5.2.1 Flash composition samples. Adequate controls to insure that the flash composition complies with the requirements shall be provided. Test for verification at least, one sample of the composition of sufficient size (see paragraph 4.7.2) as representative of each eight (8) hours production of simulators shall be provided. A composite sample shall not be used. If the sample fails to meet the requirements for granulation or composition, or if the simulator has been loaded and packed with non-conforming composition, the remaining unloaded composition shall not be used in production and the lot of loaded simulators shall be rejected.

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4.5.2.2 Granulation test. A sample of sufficient size (see paragraph 4.7.2) of each ingredient shall be selected for the granulation test at the time of loading the ingredient. A sample of sufficient size shall be taken from a "dummy" body used in conjunction with the loading of the simulators for determination of composition.

4.5.3 Functioning verifications. The simulator assemblies shall be observed for any evidence of failure to comply with the requirements, as classified in Table V.

TABLE V

Defects	Classification
Premature (See 3.5.1)	Critical
Delay time (See 3.5.1)	Major
Sound level (See 3.5.2)	Major
Pull cord failure (See 3.5.3)	Major
Audible whistle (See 3.5.4)	Major
Dud (see 3.5)	Major

4.5.3.1 Functional test for defects at ambient temperature (70° ± 5° F) (See Table V)

4.5.3.1.1 First three lots. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the requirements, 125 simulators shall be selected for this test. 75 simulator assemblies shall be functioned in the static phase and 50 simulators shall be functioned in the non-static phase. The lot shall be rejected if, during the test, any of the following are exhibited in the combined two phases.

a. One or more critical defects.

b. Six or more assemblies having major defects listed in Table V.

4.5.3.1.2 Successive lot test. After three consecutive lots have complied with the criteria of 4.5.3.1.1, fifty (50) simulators shall be selected from each lot for this test. 25 simulators shall be functioned in the static phase and 25 simulators in the non-static phase (see 4.7.4.1.2). The lot shall be rejected if, during the test, any of the below are exhibited.

a. One or more critical defects.

b. Three or more assemblies having major defects listed in Table V.

If the lot is rejected, the next lot produced will proceed as if it is the first lot being produced, and shall follow the requirements of 4.5.3.1.1 and 4.5.3.2.1 until 3 consecutive lots are accepted.

4.5.3.2 Inspection for defects at cold temperature (minus 65 + 5° F) (See Table V).

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4.5.3.2.1 First three lots. Beginning with the first lot produced and continuing until 3 consecutive lots have complied with the acceptance criteria, 80 simulators shall be selected from each lot for test. All assemblies shall be functioned in the non-static phase. The lot shall be rejected if, during the test, any of the following is exhibited.

a. One or more critical defects.

b. Four (4) or more assemblies having major defects listed in Table V.

4.5.3.2.2 Successive lot test. After three consecutive lots have complied with the criteria of 4.5.3.2.1, 50 simulators shall be selected from each lot for this test. All assemblies shall be functioned in the non-static phase. The lot shall be rejected if, during the test, any of the following are exhibited.

a. One or more critical defects.

b. Three or more assemblies having major defects listed in Table V.

If the lot is rejected, the next lot produced will proceed as if it is the first lot being produced, and shall follow the requirements of 4.5.3.1.1 and 4.5.3.2.1 until 3 consecutive lots are accepted.

TABLE VI. Destructive test.

Temp.	Initial		Reduced Sampling Plan (After 3 consecutive complying lots)	
	Number of Tests	Reject if a critical defect or any of the following occur	Number of Tests	Reject if a critical defect or any of the following occur
Ambient	125	Six (6) or more major defects	50	Three (3) or more major defects
Cold	80	Four (4) or more major defects	50	Three (3) or more major defects
			Note: If a lot fails or withheld from submission during the reduced sampling plan, the next lot will go back to initial testing until another 3 consecutive lots have complied with the acceptance criteria	

4.6 Fuse length and delay.

4.6.1 Determination of critical fuse length. For each spool of fuse cord, a sample of 30 pieces of fuse nominally three (3) inches shall be cut. Fuse cords shall be cut (10 each) from the

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beginning, middle and end of the spool with respect to the lot size. Determine the average burn rate of the samples and standard deviation of the burn rates. Critical fuse length shall be determined by adding the average burn rate to four (4) standard deviations and then multiply by five (5) seconds (critical burn time). Critical fuse length =5 sec * [Average Burn Rate + 4 * STD]

4.6.1.1 Inspection for critical fuse length. Cut fuses shall be gaged and inspected 100% plus verification level VII for compliance to the determined critical fuse length. Any fuse less than critical fuse length shall be burned to determine the burn duration. If the burn duration is less than five (5) seconds then it is a critical defect and the lot shall be rejected. If the length of the fuse cord is less than the determined critical length but not a short burn then it is a major defect.

4.7 Test methods and procedures.

4.7.1 Moisture content.

4.7.1.1 Whistle composition. The method of determining the percentage of moisture in the whistle composition shall be in accordance with MIL-P-48239, except that the drying time shall be for one hour in lieu of three hours.

4.7.1.2 Flash composition. Determination of moisture content of the flash composition shall be performed by transferring a five (5) gram sample to a tared weighing dish. Weigh the dish and the contents and then place in an oven maintained at a 212°F for three (3) hours. Cool in a desiccator and weigh. Calculate the loss in weight of the dish and contents as percentage of moisture in the sample.

4.7.1.2.1 Moisture content of flash composition. Moisture content shall be controlled as specified in 4.5.1. For verification, a sample shall be taken from the contents of a “dummy: body used in conjunction with the loading of the simulators. Moisture content shall be determined as specified above.

4.7.1.3 Alternate method (for flash composition). The moisture content shall be determined in accordance with Method T101.4 of MIL-STD-286. Timer shall be set to the minimum time required to obtain constant readings. Temperature settings shall be 212°F +/- 41°F.

4.7.2 Granulation and composition.

4.7.2.1 Granulation. The particle size shall be determined in accordance with Method A3 of ISO 10070.

In order to avoid the dangerous operation of screening the flash composition, this test may be performed by screening the individual ingredients prior to blending. Alternatively, particle size can be verified by using the individual specification which is specific to the pyrotechnic powder.

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4.7.2.2 Composition of flash powder.

4.7.2.2.1 Composition of KP/Al flash powder. The composition shall be determined in accordance with MIL-P-48239.

4.7.2.2.2 Composition of BP/Al flash powder. Transfer an accurately weighed portion of approximately 2 grams of the flash powder that was dried for the moisture determination to a tared crucible. Wash the sample between 10 and 15 times with 5 milliliter (mL) portions of cold carbon disulfide and rinse with ether. Continue washing the sample with fifteen (15), 20 mL portions of cold distilled water, allowing each wash of the water to remain in contact with the sample approximately 1 minute before applying suction. Aspirate the crucible and contents thoroughly, and dry in an oven maintained at a temperature of 100°C + 5°C. The crucible and contents shall be cooled in a desiccator and weighed. To the contents remaining in the crucible add several portions of hot, dilute hydrochloric acid until the reaction subsides. After all reaction has ceased, wash the residue with several portions of hot water. Dry the crucible and contents in the 100°C + 5°C oven and then cool in a desiccator and weigh. Calculate the percent of aluminum as follows:

$$\text{Percent aluminum} = \frac{(A - B) * 100}{W}$$

Where:

A = weight of crucible and contents after the carbon disulfide and water extraction, in grams.
B = Weight of crucible and contents after the acid extraction, in grams.
W = Weight of the sample, in grams.

Determine the percentage of black powder by subtracting the percent of aluminum from 100.0 percent.

When blending is accomplished in individual charge cases, the adequacy of the composition may be determined by emptying one or more of the partially mixed charges from the charge cases before they have been closed, blending the ingredients and performing and analysis on the sample of the mixture.

4.7.2.2.2 Flash Testing Composition (Alternate). Use of PLC controlled auto-loading machines for composition of BP/AL flash composition. Calibrate scale(s) of auto loading machines at the beginning of each shift. Dispense 3.3 grams +/-0.4 grams of aluminum into the whistle and cover assembly (DWG 7549235). Dispense 36.8 grams +/- 0.4 grams of black powder into the whistle and cover assembly (DWG 7549235). Verify accuracy of each component of the flash composition through SPC. Evaluate machine data for accuracy and include printed data in lot history files for review by the Government.

4.7.3 Jumble. The simulator assemblies shall be tested in accordance with procedures specified in MIL-STD-331, except that the duration of the test shall be limited to a minimum of 20 minutes. The assemblies and equipment shall be observed and inspected for compliance with paragraph 3.4.

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4.7.4 Conditioning requirements.

4.7.4.1 Ambient temperature - (70°F ± 5°). The simulator assemblies shall be conditioned for not less than twenty-four (24) hours at the specified ambient temperatures. At the end of the conditioning period, the assemblies shall be tested in the following manner within 30 minutes after removal from the conditioning chamber. The functioning performance shall be observed for compliance with the requirements.

4.7.4.1.1 Static. The assemblies, conditioned at ambient temperature, shall be mounted horizontally on a rigid horizontal support and fired by means of a lanyard attached to the fuse igniter cord. The assembly shall not be shaken or agitated in any manner other than normal handling within two (2) hours prior to test. Measuring of sound level intensity shall be accomplished using sound Analyzers, microphone System and Sound Level Meters and the pick-up of the sound level shall be located 75 ± 0.5 feet from the simulator test fixture.

4.7.4.1.2 Non-static. The assemblies, conditioned at ambient temperature, shall be fired by placing the assembly in an ejection machine (DWG. 9200454 or equivalent device). The fuse igniter cord shall be held in a fixed position. There shall be no slack evident in the igniter cord prior to initiation. Upon ejection of the assembly, the simulator shall be initiated. The throwing range of the ejection machine shall be not less than 45 feet and the ground shall be free of large rocks. The assembly shall be measured and observed for compliance with the requirements specified. Sound level intensity need not be measured during this test.

4.7.4.2 Cold temperature (-65°F ± 5°F). The simulator assemblies shall be conditioned for 16 hours at the required cold temperature and fired within five (5) minutes after removal from the conditioning chamber by the same method specified in 4.7.4.1.2 (Non-static). The functioning shall be measured and observed for compliance with the requirements. If for any reason the units are not able to be functioned within five (5) minutes, a cooler that has a stabilized controlled temperature may be used to transport the units to the testing site for function test.

4.8 Ammunition lot numbering. Visually verify that ammunition lot number has been applied to each lot as described in MIL-STD-1168.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES
(This section contains information of general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M115A2 simulator is used exclusively in training to imitate battle sounds and flashes. It creates battle sounds and flashes mimicking that of shells in flight and ground explosions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and of all reference documents cited in Section 2, and listed in Section 6 for information.

b. Provisions for submission of first article samples.

c. Provisions for submission of Conformance/Lot acceptance samples.

d. Packaging requirements (see 5.1 and 6.6).

e. Requirements for submission of acceptance inspection equipment (see 6.3).

f. Provisions for critical characteristic contract clause.

g. Statistical Process Control (SPC) clause.

6.3 Submission of acceptance inspection equipment. Acceptance inspection equipment designs should be submitted as required to Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: RDAR-QEM-B, Picatinny Arsenal, NJ 07806-5000.

6.4 Distribution of ammunition data cards. Distribution of data cards should include the following: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: RDAR-QEM-B, Picatinny Arsenal, NJ 07806-5000.

6.5 Justification of critical defects.

6.5.1 Short fuse delay (4.3.2.6 critical 1). If the delay is short (less than five (5) seconds) then the end user may get seriously injured. Also, same possibility that can happen as described in paragraph 6.5.5.

6.5.2 Any opening in assembly which charge may leak, or presence of charge on exterior of assembly (Assembly) (4.3.2.12 critical 1). If there are any leaks from the assembly or presence of charge on exterior of the assembly then that may result in accidental ignition or inadvertent functioning of the simulator. This will result in an unintentional function of the simulator during handling.

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6.5.3 Safety clip missing, insecurely engaged or incorrectly positioned (4.3.2.12 critical 2). If the safety clip is missing or not positioned correctly then the simulator is vulnerable for accidental ignition, since the safety clip acts as an assembly aid to help accidental ignition. The simulator will function prior to the operator being able to move to a safe operational distance.

6.5.4 Jumble (4.3.2.13 critical 1). If a simulator functions during the jumble test then it will be unsafe to handle also if powder leaks then it may cause unintentional functioning of the simulator.

6.5.5 Premature functioning (Time from initiation to detonation less than 5 seconds (4.3.2.13 critical 3). If the time from initiation to detonation is less than 5 seconds, the risk of detonating in the users hands is very high since the user has very little time to release and throw the simulator away of their hand or move to a safe distance.

6.6 Packaging requirements and inspections. Because packaging requirements and inspections are not permitted in an item specification in accordance with MIL-STD-961, these requirements and inspections are in drawing 13028657. Requirements for First Article should include both this specification and drawing 13028657.

6.7 Subject term (key word) listings.
Tube
Disc
Fuse
Flash composition
Whistle composition
Flash
Whistle

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian activity:
Army – AR
Navy – OS
Air Force - 11

Preparing Activity:
Army-AR
(Project 1370-2017-001)

Review Activities:
Navy – AS, NP, MC
Air Force – 70, 99

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